

Question 1.

You are given an array of integers: {12, 7, -5, 9, 20, -3, 15, 8}. Write a Java program to perform the following operations:

1. **Sort the array in ascending order** without using any built-in sorting functions.
2. **Find the second smallest and second largest elements** in the array.
3. **Reverse the array** and print the reversed array.
4. **Count the number of positive and negative numbers** in the array.
5. Create a new array that contains only the even numbers from the original array and print it.

Question 2.

Write a Java program to perform the following operations on a 2D array (matrix):

1. Create a 2D array of integers with user-defined rows and columns. Populate it with user input.
2. Print the transpose of the matrix.
3. Check if the matrix is a symmetric matrix (i.e., it is equal to its transpose).
4. Calculate and print the sum of all elements in the matrix.
5. Find the maximum and minimum values in the matrix and their respective positions.

Question 3.

Write a Java program to perform the following operations on an array of integers:

1. Create an array of integers using user input.
2. Separate the array into two arrays:
 - One containing **odd numbers**.
 - The other contains **even numbers**.
3. Sort both the odd and even arrays in ascending order.
4. Merge the sorted odd and even arrays back into a single array where odd numbers come first, followed by even numbers.
5. Print the final merged array.